ADC: AD7606C−16 (pin−compatible with AD7606B; symbol keeps AD7606 pin names)

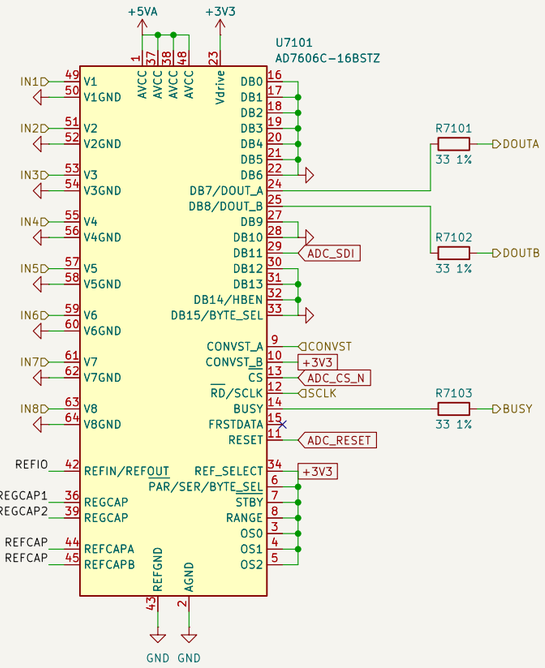
Software mode: OS0−OS2 high. FPGA writes registers over SDI (pin 29, DB11/SDI) at boot:

RANGE_CHx = 0x1 (+−5 V), BANDWIDTH = 0xFF (high bandwidth on all 8 channels), CONFIG DOUT_FORMAT = 2 lines.

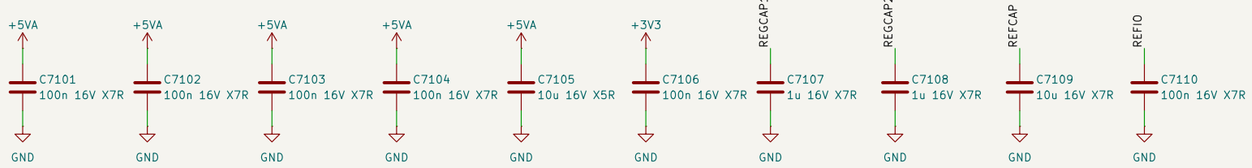
Pin 10 is WR on the B/C parts (parallel register writes only): held high. RANGE pin is ignored in software mode.

33 R source resistors on DOUTA/DOUTB/BUSY (ribbon to the FPGA); BUSY is used from ADC1 only.

DB9/DB10 are tied to GND: never set DOUT_FORMAT to 4 lines. After power−up the FPGA pulses ADC_RESET (full reset), then writes the registers.



Decoupling (place at the pins): four 100 nF on the AVCC pins + 10 uF bulk, 100 nF on VDRIVE; REGCAP 1 uF x2, REFCAP 10 uF, REFIN/OUT 100 nF per datasheet



Hybrid 8x8 ultrasonic AESA	
Sheet: /ADCs/ADC1/	
File: adc.kicad_sch	
Title: ADC (x8): AD7606C−16, software mode, serial 2 DOUT	
Size: A3	Date: 2026−10−20
KiCad E.D.A. 10.0.6	Id: 98/115
	Rev: 2